

SEDAR 51 Gulf of Mexico Gray Snapper

Data Workshop

Life History Working Group:

Stock Definition and Description

- Expand upon the genetics work of Gold et al. (2009; SEDAR51-RD-04). Gray snapper fishery-dependent sampling can be sporadic and site-specific samples are needed. SEAMAP samples may be able to help fill this void.
- Identify natal origin of adult gray snapper by using otolith chemical signatures from juveniles. Since these signatures can vary annually, samples from an ongoing sampling program would be required. Florida fishery independent monitoring program may be able to provide this for the eastern gulf.

Natural Mortality

- LHW group recommends using the maximum age of 28 years to calculate natural mortality by applying Hoenig's regression for teleosts ($M = 0.15$) as the target M used to calculate an age-specific vector of M .
- The LHW group recommends applying the age-specific mortality vector using Lorenzen (2005).
- Therefore, the LHW group recommended using a wider range of sensitivity around M (lower bound = 0.13; upper bound = 0.17).
- Review of Then et al. 2015 data inputs, possibly restrict dataset to take into consideration fishes of similar maximum size, age, and latitude/longitude to improve the estimation of natural mortality.
- Expand fishery-independent sampling

Age and Growth

- Continue annual ageing workshops with GSMFC.
- Update the gray snapper reference set to include a greater range of ages, sampling locations and section quality.
- Expand sampling of otoliths from the central and western Gulf of Mexico.

- Increase resolution of jurisdictional waters within sampling programs, specifically for the head boat program.
- Expand fishery dependent sampling.
- The LHW group recommends all gray snapper data in the Gulf of Mexico, regardless of gender, to be combined and used in SEDAR51 assessment models.
- The LHW group recommends using the predicted growth parameters from the growth model using a CV that increases linearly with age variance structure from the size limit C scenario.
- The LHW group also recommends allowing the stock assessment analyst to decide whether to predict growth within the assessment model, using the recommended growth parameters as priors, or to use the growth parameters in the assessment model as fixed parameters.

Reproduction

- Due to the lack of fecundity information the female weight-length relationship should be used to calculate spawning stock biomass. Additionally, the 300mm FL mark should be used as the size at maturity parameter because of the marked contribution to the spawning stock beyond that point.
- More at sea sampling is necessary to obtain more freshly-fixed ovaries to be used for spawning frequency and fecundity analysis.
- More ovarian samples need to be obtained from the western Gulf of Mexico, as that region is poorly represented in life history information.
- Workshops are pivotal to establish consistency in reproductive histological determination throughout the Gulf of Mexico.
- Locate and characterize spawning habitat and spawning aggregations.

Steepness

- Based on the limited information available, the LHW group recommends a range of steepness from 0.81 to 0.99 based on the results and range considered in the mutton snapper assessment. The LHW group agreed that mutton snapper would be the more

similar species to gray snapper as opposed to red snapper based on habitat (e.g., occurrence on reefs (Ault et al. 2006; Bryan et al. 2013)) and fishery characteristics.

Habitat Requirements

- Good information is available for juveniles in estuarine habitat, but data on ontogenetic shifts and habitat connectivity are needed between mangrove and seagrass habitats within estuaries to nearshore hard bottom and offshore spawning aggregations. Detailed classification of nearshore habitat is needed, possibly using video data (Low visibility for video work in nearshore habitats where gray snapper are located).
- Increase surveys in western Gulf of Mexico (Louisiana and Texas have ongoing sampling programs catching juvenile gray snapper, but information on adults lacking).
- Investigate more effective methods to capture adult gray snapper throughout the Gulf of Mexico. Develop alternative strategies for fishery-independent sampling (small hooks, night fishing, and video work).
- Expand seagrass mapping to track seagrass habitat changes over time. Along the coast of northern Gulf of Mexico observations of seagrass loss are being made in association with increasing water turbidity, whereas Tampa Bay has made significant progress in recovering seagrass habitat.
- Mangrove habitats are increasing all along the Gulf of Mexico coast, so tracking this expansion of mangrove habitat in Texas (black mangrove) and Florida (red mangrove) would be useful considering gray snapper's association with mangroves.

Movements and Migrations

- More tagging information including acoustic tagging, is needed to clarify temporal and spatial patterns of gray snapper movement, specifically movement to spawning aggregations. While there is some evidence gray snapper migrate to spawn, more information is needed to confirm these patterns.

Episodic Events

- More research should be done on the effects of episodic events on all life stages of gray snapper.
- Review available unpublished records on fish kills.

- Collections of fish during episodic events would allow for the size/age selectivity of mortality to be determined, and might also allow for some minimum estimates of total mortality.

Commercial Fishery Statistics Working Group:

- Consistent and sufficient levels of observers are needed in both the Gulf of Mexico and Monroe County, FL, to document discard length and mortality.
- Increase biological sampling efforts to better define stock boundaries.
- Automate volunteer and required fisher data reporting to reduce reporting complexity and time commitment for fishermen. Automation will improve both data quality and quantity.

Recreational Fishery Statistics Working Group:

1. Future MRIP calibration consideration – provide cell-level adjusted time-series of catch estimates (state, wave, mode, area-fished by species).
2. Explore landings and length composition by area to investigate possible species stock expansion.
3. Determine adequacy of regional sampling design given changing species distributions and developing fisheries (ex: spearfishing, targeted hook & line methods). Continue to monitor regions and gears where gray snapper observations are increasing.
Adequately sample inshore fishing in LA.

Measures of Population Abundance Working Group:

During the review and evaluation of the various datasets and indices presented during the Data Workshop, the IWG identified the following research recommendations to further improve the abundance indices:

- Further exploration of various weighting alternatives for the combined video index from NMFS Mississippi Labs, NMFS Panama City, and FWC video datasets – especially in terms of ongoing and future survey expansion (new areas, overlapping areas, new habitats such as artificial reefs)
- Continued bottom mapping and identification of new reef to estimate Q

- Move to a unified, habitat-based sampling design, including the MS, PC, and FWC into a single sampling program
- Explore combining data from FWRI long-term and polyhaline seagrass surveys
- Explore temporal stability of Stevens-MacCall, and how to address when species relationships change through time

Assessment Workshop

1. Additional analyses are needed to determine biologically plausible F_{msy} and MSY proxies.
2. Evaluate existing methods for deriving discard rates evaluate technical improvements to estimation methods as appropriate.
3. Develop/evaluate methods to maintain continuity of fishery-dependent indices in light of management regulations.
4. Identify factors resulting in the release of fish in excess of size limits and improve estimates of asymptotic retention.
5. Conduct additional research on fleet-specific discard mortality rates.

Review Workshop

The Review Workshop Panelists considered the research recommendations provided by both the Data and Assessment workshops and provide additional recommendations.

Data and Modeling Recommendations

- Continue to investigate the age and length discrepancies identified during the assessment process. This could be due to a problem in the data itself or model-misspecification (e.g. stock ID issues or regional variation in population dynamics).
- Information on release mortality rates over a broader range of depth

Longer-term Recommendations

- Improve biological sampling of catch, particularly discard information (length composition, discard rates)
- Need for data, particularly the recreational data, to be provided at a finer spatial scale

- Simulation modeling to determine direct economic value to the fishery for improvements in data sources or modeling efforts
- Habitat suitability and distribution modeling
- Improve communications and research collaborations between various groups within the SEFSC to improve data and assessment products

Recommendations on possible ways to improve the SEDAR process

- Provide the Review Panel with a description of the SEDAR Process (assessment) and the Council Process (management after the assessment is complete).
 - A short presentation/synthesis on the basic biology of the species being assessed and a description of the fishery should be provided to the Review Panel. Standard infographics could be developed for this purpose and over time, provide for all stocks/fisheries.
- It might be helpful to have one member of the SSC participate in all three stages of the assessment process (Data, Assessment and Review workshops); this would help in resolving questions about decisions that were made prior to the review workshop.
- Several members of the panel felt that embedding figures into the text of the assessment report (rather than appending them) would make it easier to follow the report.
- Agree in advance a suite of tools for model validation
- Implement parallel processing so that more scenarios can be run